

Course 6: Capstone: Applying Project Management in the Real World

Module 1-Initiating a project

Course Overview

- The course helps learners develop a portfolio of project management artifacts demonstrating skills like stakeholder management, team coordination, planning, and communication.
- It prepares learners for project management job interviews by reflecting on past projects, creating an elevator pitch, and anticipating common questions.

Course Structure and Activities

- The course includes videos, readings, quizzes, peer reviews, and hands-on activities such as analyzing project documents, completing project charters, and defining quality standards.
- Learners observe a project manager in action and practice tasks like identifying project requirements, prioritizing tasks, and reporting project impact.

Learning Support and Flexibility

- Google project managers provide instruction, strategies, and resources throughout the course.
- Learners can progress at their own pace with personalized deadlines and access to a private community for networking and support.

Recommendations for Success

- It is recommended to complete the previous courses in the program before starting this capstone to build a strong foundation.
- Engaging fully with all learning materials and activities enhances understanding and skill mastery.

Purpose of the Course

- Develop a portfolio of project management documents demonstrating skills in managing stakeholders, organizing plans, and communicating project details.
- Apply learned concepts by building project artifacts relevant to real-world projects or job applications.

Creating and Organizing Artifacts

- Engage in activities that involve creating project management documents throughout the course.
- Save templates and completed artifacts in a digital folder or online storage for easy access and sharing.

Career Readiness

- Use the portfolio of artifacts to showcase project management skills to future employers or team members.
- Organizing work in one place facilitates sharing and demonstrates preparedness for project management roles.

- Analyze project documents and supporting materials to identify project requirements, evaluate stakeholders, and problem-solve.
- Use a project charter as a tool to align project scope and goals among stakeholders.
- Apply effective negotiation skills with stakeholders to prioritize project goals.

Strategic thinking involves analyzing documentation and talking with stakeholders to inform decisions based on the information available to you.

Project Charter

Project charter: A formal document that clearly defines the project and outlines the necessary details to reach the project's goals.

A project charter:

- Helps organize vital project information
- Helps create a framework for the work that needs to be done
- Helps communicate those details to the necessary people
- Is useful as a reference

The goal of the summary is to provide an overview of the project and to outline the goals you hope to accomplish.

Project goals: Refer to the desired results of the project. They are determined by input from stakeholders and the project manager. Project goals address the overall result that stakeholders are aiming to achieve.

Deliverables: Refer to the specific tasks and tangible outcomes that enable the team to meet project goals.

Example:

Goal: Launching the tablet rollout project

Deliverable: Successful installation of working tablets at two restaurant locations

The project charter contains:

- Summary
- Goals
- Deliverables
- Scope

Out-of-scope: Details that don't contribute to the project's goals.

The primary components of a project charter: Summary, Goals, Deliverables, Scope. Discuss how to use the charter as a tool to get the project's stakeholders in alignment about the project's scope and goals.

Stakeholder Alignment Considerations

- Who are your stakeholders?
- What details are most important to them?
- Are there project details that your stakeholders might disagree with?

A project charter is also an alignment tool.

Alignment: Reaching agreement between two or more parties.

A common cause of project failure is misalignment among stakeholders about the details of the project. Misalignment can also happen between you and your stakeholders when you have differing visions for the project.

- Align with your stakeholders before the work begins.
- Take time during the initiation phase to create a project charter that clearly lays out the details of a project.

When presenting a project charter to stakeholders:

- Collect feedback
- Identify where there are misalignments
- Make changes to address those misalignments

Documenting misalignments and their resolutions in the project charter allows you to reference those decisions later on. Think about your audience and what information is important to them. Document misalignments and their resolutions in your notes and the project charter to reference later on.

SMART Goals and Their Importance

- Project goals should be clear and specific to avoid misalignment and to clarify the project's scope, budget, and timeline.
- Specific goals help determine project deliverables, which are the tasks or outcomes needed to achieve those goals.

Making Goals SMART

- The SMART method ensures goals are Specific, Measurable, Attainable, Relevant, and Time-bound.
- Specificity involves defining what exactly is to be accomplished and clarifying subjective terms by consulting stakeholders.

Measuring and Aligning Goals

- Measurable goals include quantifiable targets, often established through benchmarking industry success metrics.
- Goals must be attainable with available resources, relevant to the organization's broader objectives, and have clear deadlines to track progress effectively.

Goals should be: **S**pecific, **M**easurable, **A**ttainable, **R**elevant, **T**ime-bound.

Scope Definition and Stakeholder Alignment

- **Scope:** Defines the boundaries of the project, specifying what is included and excluded to achieve project goals.
- **Out-of-scope:** Details that are not included in the project plan and don't contribute to the project's goal.
- It is important to identify which elements stakeholders agree are in-scope or out-of-scope and resolve any disagreements early.
- Questions to ask: Which project details are your stakeholders aligned on that would be considered in-scope? Do your stakeholders disagree on any elements? Are there any details you should designate as out-of-scope for this particular project?

Benefits and Costs in the Project Charter

- **Benefits:** The expected gains of a project, which can be monetary or indirect, such as improved customer satisfaction. A list of benefits can help you identify potential project goals you might have missed.
- **Budget:** An estimate of the amount of money allocated to complete the project.
- **Costs:** Include money spent on labor, materials, time, and resources, typically managed through a project budget. A list of costs will include items the organization will have to pay for in order to get the job done.
- The benefits of a project should outweigh the costs.

Summary of core elements: **Scope** (Boundaries of the project), **Benefits** (Expected gains of the project), **Costs** (Money spent on the project).

Using the Charter to Communicate and Gain Consensus

- Emphasizing key benefits relevant to stakeholders helps gain their support.
- Listing benefits and costs in the charter makes it easier for stakeholders to weigh the value of the project and agree on its worthiness before detailed plans are finalized.

Key Challenges and Strategies in Project Initiation

- **Challenges:** Aligning key stakeholders with different perspectives is crucial to avoid conflicts and ensure project success. Attracting talented project managers through incentives is more effective than simply assigning tasks. Common reasons for project failure include poor project definition, lack of organizational framework, and having too many people involved without clear accountability. Managing diverse stakeholder perspectives requires frequent communication and compromise to form a solid project foundation.
- **Strategies:** Asking detailed questions about the project environment, risks, and measurable factors helps refine success criteria. Conducting surveys and interviews with people in the environment provides insights into potential problems.

Stakeholder Analysis

Stakeholder analysis: A visual representation and mapping of all of the stakeholders, their priorities, and communication preferences.

A stakeholder analysis helps you understand:

- Who your stakeholders are
- What their priorities are
- How they like to communicate
- How to prioritize stakeholder needs and goals

It helps identify which stakeholders have high influence and interest, guiding prioritization and negotiation strategies.

- **Influence:** Related to how much power stakeholders have, and how much their actions affect the project outcome.
- **Interest:** Refers to how much the stakeholder's needs will be affected by project operations and outcomes.

Stakeholders with high influence and interest should be kept informed and their goals prioritized. Mapping stakeholders on a power-interest grid clarifies whom to engage closely and how to approach negotiations effectively.

Negotiation Topics and Constraints

Common project negotiation areas include:

- Scope
- Costs and benefits
- Timeline
- Success criteria
- Roles and responsibilities
- Resources

Early alignment with stakeholders, especially on scope, can reduce conflicts and facilitate smoother negotiations later.

Stakeholder Management Importance

- Maintaining good relationships with influential people in a project encourages communication, teamwork, and trust. Effective stakeholder management increases the likelihood of project success.
- **During Initiation:** Identify all stakeholders early to set clear expectations, responsibilities, and boundaries. Establish a clear project vision agreed upon by stakeholders to prevent scope creep and align goals.
- **Throughout Life Cycle:** Understand stakeholders' priorities, roles, and communication preferences to keep them engaged. Use senior stakeholders' support when needed and involve stakeholders in problem-solving to maintain their vested interest.

Negotiation and Persuasion in Project Management

- Persuasion and negotiation are constructive tools you can use to enhance communication, clarify wants and needs, and achieve workable solutions for everyone involved.
- Viewing stakeholders as partners rather than opponents helps build consensus and deescalate conflicts. View the people you're negotiating with as your colleagues and peers, not as opponents.
- **Mutual benefit:** When all parties involved gain some kind of benefit or advantage. The goal is to reach a solution that maximizes benefits, minimizes losses, and is fair for all. Examples include compromises like adjusting team size with automation or timeline changes to satisfy different stakeholder preferences.
- You need a clear understanding of project priorities. The triple constraint model (iron triangle) helps evaluate the impact of changes and tradeoffs across scope, timeline, and budget.

Understanding Stakeholder Motivations

- Stakeholders may want to adjust scope due to budget concerns, workload, timeline, or personal career goals.
- Recognizing these motivations helps in finding compromises that satisfy all parties.

Facilitating Scope Discussions & Negotiation Strategies

- Begin by clarifying the purpose of the meeting and shared project goals.

- Listen actively to stakeholders' concerns before presenting your views to foster understanding.
- Ask detailed questions to define specific, measurable goals.
- Explain the reasons behind your requests before stating what you want.
- Allow time for stakeholders to process information without overselling.
- Encourage creative solutions and keep discussions objective, aiming for win-win outcomes.
- **Best practices for mutually beneficial agreements:** Share information openly to communicate worries and preferences clearly, avoiding withholding details. Ask questions and listen actively to understand the other party's concerns and expectations.
- **Techniques:** Propose multiple options rather than a single solution to increase chances of finding a workable agreement. If proposals are rejected, ask which option is preferred to guide toward a mutually acceptable solution.

Four Steps to Effective Influencing (Conger's Framework)

1. Establish credibility (boosts your credibility)
2. Frame for common ground (find common ground)
3. Provide evidence
4. Connect emotionally

These steps help build trust and alignment with stakeholders during negotiations. The ability to influence is a big part of project management.

Forming a Coalition

Coalition: A temporary alliance or partnering of individuals or groups in order to achieve a common purpose or to engage in a joint activity.

- When two or more people advocate together for an idea, they're able to exert more influence than if they attempted to act alone.
- An effective coalition includes people with the right mix of influence or power and interest. Use stakeholder analysis to identify suitable coalition members and reach out with clear, persuasive communication explaining the issue and their potential contribution.
- **Persuasive coalition request components:** Clearly state the issue; Ask the person for their support; Refer to the sources of power you identified for that person.

Module 2-Building out a project plan

Purpose and Function of a Project Plan

Project plan: Helps document the scope, tasks, milestones, budget, and overall activities in order to keep the project on track.

The project schedule is central, guiding time estimates, milestones, and progress tracking.

Identifying Tasks and Milestones

- Start by reviewing the project charter's goals and deliverables.
- **Tasks:** Activities that need to be accomplished within a set period of time and assigned to team members.
- **Milestones:** Important points within the schedule that indicate progress, often indicating the completion of deliverables or phases.

Core workflow question: What steps do we need to take in order to achieve each deliverable?

Using Documentation to Uncover Tasks

Search for useful information from existing project documentation, such as the project charter, email threads, and old project plans from a similar project. Take note of information that suggests additional tasks and milestones that your team will need to complete in order to meet project deliverables.

- **Emails:** Ask to have relevant emails that contain discussions about the project's details forwarded to you. Identify team members to connect with further if you have additional questions.
- **Previous project plans:** Ask a colleague with experience launching other products for the same company to share their project plan as an example. Ask colleagues about unrelated projects that also had similar components. They provide helpful inspiration as you create your own list of tasks, and help identify possible task durations, subject matter experts, and suppliers.

Examples of tasks: Produce multiple drafts of marketing materials, Generate an email list, Program emails to be sent on correct dates.

Reviewing and Breaking Down Tasks

- Look for large tasks that can be divided into smaller ones. Is there a large task being worked on by many people that could be broken into smaller tasks assigned to individuals?
- Identify dependencies, such as prerequisite tasks needed before others can start. Are there signals that imply prior tasks need to be completed first?

Task Definition Guidelines & Team Input

- Write project task descriptions in one or two sentences to keep them clear and concise. If a task description is too long, it may need to be broken down into smaller subtasks or clarified further.
- Analyze task dependencies to understand what must be completed before starting a task, helping decide how detailed the task breakdown should be.
- Involve team members in breaking down tasks to gain different perspectives and leverage their experience.
- Define tasks based on the expected time to complete them; lengthy tasks may need further subdivision and help with scheduling and milestone setting.

- Identify what it means for a task to be "done" to ensure all necessary steps are included and to establish checkpoints for progress.

Importance of Domain Knowledge & Online Research

Domain knowledge: Knowledge of a specific industry, topic, or activity. Gaining this knowledge helps identify relevant tasks and monitor project progress effectively. Knowing where to find useful information can help you increase your domain knowledge.

Online research can help increase your knowledge of industry terms, techniques, processes, and more. Tips for project research:

- Search online for news coverage or articles about similar projects at other companies to learn about outcomes and challenges.
- Search online for research on topics related to your project using keywords and tags like "best practices".
- Research similar projects in other industries to gather ideas and approaches.
- Review your list of tasks and research the specifics of executing that work to uncover additional subtasks needed for successful execution.

Discussions with Team and Stakeholders

Discussions with other people working on the project—from stakeholders to team members—can help you uncover tasks that you're still missing or clarify the smaller subtasks. Conversations often contain more details and information than you need to create a thorough list of tasks.

Ways to discover tasks through conversations:

- **Group brainstorming:** Brainstorm as a group with team members who will likely work on those tasks or related tasks to reveal potential challenges and overlooked items.
- **One-on-one conversations:** Hold one-on-one conversations with team members about tasks they'll likely be responsible for completing to clarify responsibilities and uncover detailed tasks.
- **Leverage expertise:** Consult with other people in your organization who are experts on given tasks, even if they are not directly involved, to discover what you don't know and fill in knowledge gaps.
- **Stakeholder conversations:** Have a conversation with your key stakeholders to fill in any gaps. Stakeholders who have high- or medium-level interest or influence in the project are most likely to provide the information you need.

Prepare for stakeholder conversations: Gather information ahead of time; Outline clear questions you need answers to; Present your research and your current list of tasks.

The right level of detail to include in your task list will vary from project to project and team to team. A key part of your role is identifying the right level of detail and synthesizing it into tasks in a clear, concise project plan.

Ordering Project Tasks and Determining Milestones

Review your task list to break down any large tasks into smaller subtasks and ensure all necessary tasks are included. Finalize and order your task list based on dependencies and logical sequence.

Determining task priority / order:

- Basic order of operations / natural task sequence (Rearrange tasks in the order in which your team needs to complete them).

- Dependencies or prerequisites for each task.

Determining milestones: Milestones mark important points in the project schedule, indicating the completion of deliverables or phases.

- Identify points in the project plan where you and your team can evaluate or test the work completed so far.
- Identify important tasks that your stakeholders have a particular interest in.
- Identify tasks that carry a high risk or signal the completion of a phase or major task.

Time Estimation

Time estimation: A prediction of the total amount of time required to complete a task. Providing estimates for each task gives you a big-picture sense of the project timeline in relation to individual deadlines and milestones. Accurate estimates allow tracking progress and making timely adjustments to keep the project on schedule. Asking the right questions can help your experts get to the most accurate time estimates possible.

Effort vs. Total Duration:

- An **effort estimate** only takes into account the actual work time it takes to complete a task.
- A **total duration estimate** accounts for the effort estimation and any other factors, like getting approvals, prep work, and testing.

Strategies for getting accurate estimates:

- Check their understanding of the task; verify the expert's understanding by having them explain detailed task steps.
- Ask for estimates of the substeps and make note of them to compare totals.
- Discuss the assumptions the expert might be making when they give you an estimate (such as equipment, supplies, team size, and skill levels), and consider their impact if they change.
- Ask the task expert to consider how likely it might be that the assumptions might not work out and how that might impact their estimate.
- Compare the expert's estimates against the actual time spent on similar tasks in previous work/projects to refine accuracy.

Three-Point Estimating Technique

Three-point estimating: Used to help determine the most realistic time estimate for a task. It accounts for potential risks affecting time, budget, and resources by utilizing calculations based on the best-case, normal-case, and worst-case scenarios. Each task receives three estimates:

- **Optimistic:** Assumes the best-case scenario that issues will not occur, materials are ready, and completion is timely.
- **Most Likely:** Assumes some minor issues might occur and typical delays based on normal circumstances (how long the task usually takes).
- **Pessimistic:** Assumes the worst-case scenario—that issues will definitely occur, causing significant delays and problems.

Example Task: Training the staff to use tablets

- **Optimistic (4 hours):** Vendor is well-qualified, has all materials, and shows up on time; staff shows up on time and completes training; all equipment works.

- **Most Likely (6 hours):** Vendor is qualified but might lack some materials; vendor is new and needs extra prep time; extra training needed for absentee staff; minor equipment glitches.
- **Pessimistic (6 hours + 1 week):** Original vendor quits and a new one must be hired; staff no-shows or turnover happens right before training; equipment isn't delivered on time or doesn't work, causing delays until replacements arrive.

When conducting research or talking with experts, be mindful of these three points to determine outcomes. Document conditions for each estimate to clarify assumptions. Comparing these estimates helps build a realistic schedule with buffers for risks without being overly conservative.

Determining a final estimate: Examine best- and worst-case scenario timing and compare with the most likely scenario. Build in a "buffer" that accounts for risks that are likely, but still keeps the project progressing at an efficient rate. Choose an estimate closer to pessimistic if uncertain, or optimistic if conditions are highly favorable.

Estimation Formulas:

- *Triangular Distribution:* Averages the three estimates equally.
- *Beta (PERT) Distribution:* Weights the most likely estimate more heavily for greater accuracy, often preferred for time and cost estimates.

Confidence Level Ratings

Confidence level rating: Indicates how confident you are in an estimate's accuracy, ranging from high (very confident) to low (not confident). Estimating is inherently uncertain, especially in project management with many unknowns. Using confidence ratings and communicating uncertainty to stakeholders leads to a more accurate and realistic project timeline and better decision-making.

Ways to determine a confidence rating:

- **Poll your team:** Get a percentage for how confident they are on tasks they're assigned to and come to a consensus; define categories.
- **Use the three-point technique:** Considering best- and worst-case scenarios can increase confidence in estimates. Categorize experience with similar projects to help assign an overall confidence rating.

Adding notes about risks or uncertainties alongside confidence ratings helps identify tasks that may need closer monitoring or team input. Implementing a system that accounts for unknowns and ensures an accurate picture of project costs means everyone will be better off over the long term.

Negotiation Techniques for Time Estimates

Negotiations about time estimates differ from scope negotiations; the goal is to reach an objectively accurate estimate with task experts. People may unintentionally overestimate or underestimate time due to optimism, a desire to please, or caution.

Techniques specific to negotiating a time estimate:

- **Say "no" without saying "no":** Avoid direct refusals; use open-ended questions to encourage collaboration on alternative solutions. Ask open-ended questions like: *"How would you like me to proceed?"*, *"How can we solve this problem?"*, *"What can I do to help?"*
- **Focus on interests, not positions:** Try to identify the other person's needs, wants, and motivations around completing a certain task to find acceptable compromises.

- **Present mutually beneficial options:** Explore solutions that satisfy both parties, such as providing missing resources to reduce time estimates.
- **Insist on using objective criteria:** Base criteria on neutral, independent data like market value, research findings, previously-documented experience, and laws and regulations. Agree in advance about which objective criteria to consult and then use that information to determine your estimates, reducing reliance on raw instincts.

Understanding Empathy in Communication

Empathy: The ability to understand and feel what others are feeling or experiencing by imagining yourself in their position. Approaching conversations with empathy helps make difficult discussions, like questioning delays or time estimates, easier and more constructive. It helps in understanding different work and communication styles among team members and stakeholders, enabling tailored communication to ensure project goals and impacts are conveyed effectively.

Micromanaging: When a manager too closely observes, controls, or continuously reminds the people they're managing of the work they've been assigned (to be avoided; practice empathy instead).

Ways to bring empathy to your conversations:

- **Listen with curiosity:** Ask questions rather than making assumptions or immediate suggestions.
- **Repeat what you think you heard:** Repeat what you think the other person said to confirm understanding and encourage perspective sharing.
- **Connect with their experience:** Share your own experiences with time estimation challenges to connect and show understanding.
- **Recognize judgments:** Recognize and manage your own internal judgments to avoid negative body language or tone that can be perceived as distrust.
- **Recognize buffering:** Ask about any buffers added to time estimates to understand potential external factors affecting timelines.
- **Avoid distractions:** Avoid distractions during conversations to show full attention and respect for the other person's input.

These empathy practices help foster trust, improve communication, and support more accurate and collaborative project planning. *Handling Team Challenges Example:* Missed deadlines were addressed by understanding personal issues affecting a team member. Adjusting timelines and reallocating resources can help accommodate team members' situations.

When negotiating task estimates, asking many questions, gathering information from experienced team members, and reviewing past similar projects and timelines is crucial to making informed decisions.

Module 3-Maintaining quality

Quality Management Importance

Quality management: Involves planning and maintaining standards to ensure project success and desired outcomes. It helps reduce costs by minimizing errors and promotes collaboration and continuous feedback within the project team.

As a project manager, you're responsible for the planning and execution of the project, as well as for the successful completion of the project. Quality means making sure that you deliver what you say you will, and that you do it as efficiently as you can.

Benefits of quality management in a project: Delivering a quality product, Decreasing overhead, Increasing collaboration and ongoing reviews/evaluation.

Key Quality Management Concepts

- **Quality planning:** The process that the project manager or the team establishes and follows for identifying and determining exactly which standards of quality are relevant to the project as a whole, and how to satisfy them.
- **Quality standards:** Requirements and specifications that your product or service must meet in order to be considered successful by your organization and the customer. Tracked throughout the project lifecycle by setting standards for tasks, milestones, and deliverables.
- **Quality assurance (QA):** Consists of reviewing processes to evaluate whether or not your project is delivering an acceptable level of quality (confirming the project is on track for quality delivery). Examples: Beta testing, Internal checklists, Feedback surveys.
- **Quality control (QC):** Techniques and operational activities used to maintain quality standards and address problems when they arise.

Quality Management Plan

Quality management plan: A comprehensive document that documents all the information needed to effectively manage project quality throughout the project life cycle. It defines the policies, processes, and criteria for project quality, as well as roles and responsibilities for carrying out the plan. Aligns with overall project planning and supports achieving project goals and stakeholder expectations.

Components include: Quality Standards, Evaluation Questions, Feedback Surveys.

Establishing Quality Standards

Quality standards are specific requirements a product or service must meet to be successful.

Resources to consult for determining quality standards:

- **Project documents:** Business case, Project charter
- Conversations with experts and stakeholders
- Industry research

Common quality standard categories: Functionality, Design, Safety, Ease of use, Productivity, Effectiveness, Customer satisfaction.

It's important that your standards are objective and measurable so you can clearly identify that the standard has been met. Use your critical thinking skills to determine the right standards and adjust them

if necessary to meet the specific needs of your project. Asking detailed questions helps translate general quality concepts (like ease of use or productivity) into specific, measurable criteria. Quality standards guide testing and evaluation during planning, execution, and after project launch. If standards are not met, project plans can be adjusted to improve quality and meet stakeholder expectations.

Importance of Evaluation in Quality Management

Evaluation: Involves observing, measuring, and then comparing your findings to a set of agreed-upon criteria like quality standards or project goals. Reasons to use evaluation: To improve, To judge, To learn. Evaluation promotes learning, informs decisions, and provides accountability. It helps improve project processes, judge effectiveness, and identify unintended problems or impacts.

Evaluation Process Steps:

1. Determine the reasons for evaluating (clarify why you are evaluating, which shapes the questions you ask). Connect evaluation focus to project and organizational goals to narrow down what to assess (e.g., installing tablets, integrating tablets with the POS system, training the staff to use tablets).
2. Write out evaluation questions.

This foundation prepares you to develop evaluation indicators and apply evaluation techniques to measure project success and guide future decisions.

Creating Evaluation Questions

Evaluation question: A key question about the outcomes, impact, and/or effectiveness of your project, program, or a key aspect of your project.

Two main categories of evaluation questions:

- **How to improve (Questions for improvement):** Help you optimize ongoing workflows. *Examples:* How can we improve? What is working and what's not working? Which goals are being met? Who is benefiting? What are the most common participant reactions?
- **Help you measure and compare (Questions for measurement/comparison):** *Examples:* What were the results? Were there unintended outcomes? What were the costs and benefits? Are there any lessons to be learned? Should we continue?

Criteria for effective evaluation questions: Address stakeholder or user values, interests, and concerns; Relate to the purpose of the project and of evaluation; Are worth answering and are important for the project and beyond; Are practical and feasible to answer with available resources.

Purpose of Evaluation Indicators

Evaluation indicators: Reveal and specify the exact type of data that needs to be collected to help you answer your evaluation questions. Indicators provide pathways for answering evaluation questions, taking your question and determining the specific type of response or measurable evidence you're aiming for (such as quantities, satisfaction levels, preferences, or demographics).

- Indicators add specificity to evaluation questions, similar to how quality standards define deliverables. They help assess whether a project meets agreed-upon quality standards by providing measurable evidence.
- *Example:* To evaluate if tablets improve staff performance, indicators might include table turnover rates, tip averages, or customer ratings. Indicators can be observable signs like test scores, attendance, or behavior changes that point to achieving project goals.

Survey Purpose and Development Process

Surveys: Tools you can use to evaluate, measure, and collect data regarding the quality of a project process, goal, or deliverable. Being able to develop a survey and write survey questions is important because it demonstrates your ability to understand the goals of your project and assess how your stakeholders and users value the project. Surveys help you understand what's working and what's not working, assess the criteria you want to evaluate, and provide data that points out whether you've met your quality standards, guiding necessary adjustments.

Survey Development Process Steps:

1. Develop evaluation questions.
2. Define your evaluation indicators.
3. Determine what type of survey to design and questions to ask.

A survey question is designed to collect data which can help you answer your evaluation questions.

Examples: "Are the tablets easy to use?", "Was there enough time during the training to practice and ask questions?", "On average, how many of your side work tasks are you able to complete during a shift?", "Since using the tablets, how often have you sent back an incorrect order?"

Types of Survey Questions:

- **Open-ended:** Require more than one-word answers such as yes or no (require detailed, descriptive responses).
- **Closed-ended:** Can be answered with a single response (yes/no, multiple choice, or scaled rating).

Tips for creating good survey questions:

- Be careful not to make assumptions about your respondents.
- Make sure your questions ask exactly what you mean to ask.
- Each question should be specific and address only one measurable aspect.
- Make sure your questions don't provide too much detail or leading information that could bias answers.

Survey data helps you determine if you're achieving your quality goals and where you need to make adjustments.

Determining Your Audience & Choosing Reporting Styles

- **Determining Your Audience:** Consider who will receive the evaluation report and what information is most meaningful to them. Different audiences (team members, managers, stakeholders, executives) may require different levels of detail, time commitments, and presentation styles. Present the same data in different ways based on their needs.
- **Choosing Reporting Styles:** Create a detailed evaluation report addressing all evaluation questions. Summarize the information into the most appropriate format for a given audience. Common reporting styles include:
 - *Full detailed report:* Comprehensive document with background details.
 - *Summary sheet:* One- or two-page snapshots or snapshots.
 - *Slide-based presentation:* Tailored for quick, structured visual delivery.

Presenting Evaluation Findings & Storytelling

An evaluation is more than raw data; it needs to reflect what the data means and explains how data responds to evaluation questions. Go beyond raw data by filtering and analyzing to become familiar with the results, the respondents, and what those results mean in regards to project quality.

Storytelling: The process of turning facts into a narrative to communicate something clearly to your audience. Shape the story of your findings by tying it all together into one cohesive narrative that highlights key findings, major issues, and recommendations, making the data understandable and actionable for stakeholders.

Evaluation Presentation Structure and Content:

- **Introduction:** Summarizes the project goals, outcomes, key findings, lessons learned, and recommendations briefly. States what is being evaluated, why, and the quality standards and evaluation questions used. Start with an engaging hook to maintain stakeholder interest.
- **Findings Body:** Each quality standard is addressed separately with a summary of findings and their implications. Data should be interpreted to tell a story that helps stakeholders understand project success or issues, supported by clear visuals like charts. Use slide-based presentations with concise bullet points and visuals, tailoring communication strictly to the audience.
- **Conclusion:** Restates findings and offers actionable recommendations for future project phases or subsequent projects.

Core workflow parameters for reporting: Determining your audience, Choosing the style of your evaluation report, Delivering an effective presentation, Distinguishing between reviewing data and presenting an evaluation.

Definition and Purpose of Retrospectives

Retrospectives: A workshop or meeting that gives project teams dedicated time to reflect on a project. Conducted at the end of projects to review what went well, what went poorly, and where luck played a role. They aim to capture lessons learned to improve future projects, but can also be conducted mid-project after major milestones when key decisions need to be made to improve ongoing processes.

Three main purposes / benefits of retrospectives:

- Encourage team building by fostering understanding of different perspectives.
- Facilitate improved collaboration.
- Promote positive changes in future procedures and workflows.

Retrospectives should be positive and blameless, focusing on continuous improvement of individuals, teams, and processes. They involve gathering input from all project participants, especially those who will work on future projects. A common failure is low participation, often due to either indifference or a lack of psychological safety. Low participation during a retrospective can prevent a project team from making meaningful process improvements.

Role of the Project Manager in Retrospectives: Responsible for managing the meeting agenda and tone to ensure inclusivity and respect. Because a retrospective is a specific type of meeting, it's crucial to have an agenda to help guide the discussion, organize the meeting, and document learnings. It's the project manager's role to facilitate a respectful and productive retrospective discussion that recognizes successes and areas for improvement, documenting key details for future reference.

Creating a Safe Environment & Modeling Participation

- **Establish a Safe Environment:** Establish a policy that ensures confidentiality, such as *"what's said here stays here, what's learned here leaves,"* to make the team feel safe. Emphasize that retrospectives are free from external stakeholders or customers, providing an open space for honest discussion about challenges.
- **Model Participation:** The facilitator/project manager should model the kind of participation they'd like to elicit from their team. Start by sharing your own successes and challenges to set the tone; admitting personal mistakes encourages team members to openly share their own experiences.
- **Techniques to Encourage Engagement:**
 - Pose a group question and ask for individual responses (require each member to share one success and one challenge to prevent one voice from dominating).
 - If initial questions don't work, try alternative phrasing like: *"What about this project should we start, stop, and continue?"*
 - Review the project timeline: Reviewing the timeline can help team members recall and discuss events from the entire project lifecycle, not just recent ones.

Accountability vs. Blame

Accountability and blame are two very different things, and only accountability belongs in a retrospective. Blame is counterproductive as it shuts down honest sharing and should be avoided.

Accountability: Being responsible for decisions associated with a project or task. It encourages the team to think holistically about mistakes and challenges, identifies solutions for the future, and encourages active ownership.

Techniques to encourage accountability during a retrospective:

- Come prepared with specific challenges to discuss as a group, especially to balance discussions that focus only on successes.
- Turn team complaints into SMART action items to promote solution-oriented thinking. SMART action items are Specific, Measurable, Attainable, Relevant, and Time-bound.
- Push the team to identify its role in creating a given challenge, even when issues seem initially outside their control, to foster insights and process improvements.
- Detach the challenge being discussed from any specific person in the room; ensure criticism remains constructive (Constructive criticism is a respectful form of feedback that is intended to help the recipient improve a piece of work) and redirect conversations away from blaming individuals toward process improvements.

Setting a Positive Tone and Managing Negativity

- **Setting a Positive Tone:** Aim to set a positive tone at the start of the meeting. Determine how you'll set the tone of the meeting, such as highlighting project successes or thanking the team for milestones achieved. Use meeting props like green and red index cards to encourage sharing both successes and challenges. Ask yourself: *Is this conversation likely to feel stressful for the team?* Retrospectives are a great way to build trust, honesty, and direct communication.
- **Understanding Negativity:** Recognize that negativity often arises if the team does not feel psychologically safe, or if individuals have underlying stressors. Ask yourself: *Does this person feel insecure about the value they add to this team? Does this person receive negative feedback on the quality of their work?*

- **Techniques for addressing negativity:** Try anticipating potential negativity by meeting one-on-one with team members before the retrospective happens to understand potential sources of negativity and reassure insecure members. Consider asking team members individually to share their thoughts to prevent one negative voice from dominating. If tension escalates, call a meeting break or timeout to deescalate tension and reset the meeting tone.

Module 4-Effective stakeholder communication

Communicating Project Problems

Every project has its problems, and communicating those problems clearly is a core part of your job as a project manager. Problems are usually handled within the project team, but sometimes they require escalation to senior stakeholders or sponsors for input, decisions, and guidance.

Synthesizing Information

Project managers synthesize relevant information from multiple sources into a coherent summary that clearly communicates the issue. Synthesizing requires gathering information from multiple sources (such as emails, presentations, meeting notes, trackers, and charters) and combining these points to help form your own coherent analysis rather than just sharing raw documents or unorganized links. This skill is essential for leading conversations, meetings, and writing clear project documentation.

Writing a Problem Summary

A problem summary should be brief (typically a one- to two-sentence summary) that clearly states the problem and its impact on the project or timeline. It should also immediately propose a structured solution or recommendation to help stakeholders make decisions quickly.

Example: "A number of tasks have run past their due date because of supplier issues, so we recommend hiring a second supplier to hit the deliverable date. Otherwise, we'll need to push the launch date back."

The Role of Problem Solving in Program Management

Problem solving is the core responsibility of program managers, involving addressing complex issues related to scope, budget, personnel, and timelines. The process includes identifying the root cause of problems, which are often hidden beneath surface symptoms, and developing structured, strategic solutions. Program managers use frameworks, tools, processes, and methodologies to organize and propose solutions effectively. Documentation such as charters, scope documents, meeting notes, and trackers are tools to facilitate managing programs, but are not the end goal themselves. Developing problem-solving expertise requires practice, which can be gained through various activities like hobbies, software projects, or work in different industries. The ability to solve complex, strategic, and holistic problems is a key skill for program managers, especially in large organizations like Google.

Objectives and Key Results (OKRs)

Objectives and Key Results (OKRs): A tool for organization-wide goal setting that combines an objective (what to achieve) with key results (how success is measured and tracked).

Example: An objective might be to prioritize customer needs, with a key result being responding to customer feedback within 24 hours.

- Projects aligned with an organization's OKRs are more likely to be relevant, strategic, and worth the investment. OKRs serve as a shared language within an organization, helping to evaluate and justify project goals.
- **Communicating Problems Using OKRs:** Linking project problems to their direct impact on organizational OKRs helps communicate the urgency and risk to stakeholders. OKRs can be really helpful to reference when communicating with stakeholders about project problems because this connection captures stakeholders' attention by showing how addressing the problem supports

broader organizational goals and impacts the business as a whole. Connect problem statements to OKRs effectively to enhance stakeholder communication relevance.

Email Writing Best Practices for Stakeholder Decisions

Email can be an effective tool for quickly escalating a problem and asking for a decision from a stakeholder on how to proceed. Helpful tips for writing strong, professional emails:

- Think about what's most important to your stakeholder.
- Identify how a problem will impact the organization as a whole and ensure you clearly communicate that impact on organizational goals/OKRs within the first two sentences of your email.
- Write a clear, descriptive subject line that states the email's purpose and indicates what you'd like your stakeholder to do upon reading your email (use tags or explicit language like "please review", "action required", or "urgent").
- Keep the body of the email brief, clear, and strictly to the point, outlining the problem, its impact on OKRs, and the specific decision or action needed. Use concise language, bullet points, and highlighted key information (including hard deadlines) to make it easy for stakeholders to digest and respond.
- Consider including attachments or hyperlinks to relevant supporting documentation in your email to ensure informed decision-making.
- Always proofread thoroughly for misspellings, grammatical errors, and inaccurate or broken hyperlinks before sending to ensure maximum professionalism and absolute clarity.

Using Generative AI for Project Communication

- Generative AI can help generate a list of necessary communications such as project status reports, communication plans, and stakeholder updates to keep a project running smoothly.
- AI tools can format outputs in organized tables or structured bullet lists, helping organize communications and ensuring no important operational information is missed.
- It is important to specify the desired output format when prompting AI tools. Project managers should align AI-generated communications with company templates, communication norms, and stakeholder expectations.
- AI can assist in drafting communications like status emails to senior sponsors, saving substantial time and effort.
- **Crucial Privacy Note:** Always review company policies and privacy terms before including any sensitive, proprietary, or confidential project information in AI prompts to protect organizational privacy.

Understanding Stakeholder Communication Preferences

- Identify the preferred communication style of each stakeholder by consulting with their executive assistants, colleagues, or past documentation.
- Recognize that different stakeholders require different levels of detail, frequency, and formats. Anticipate key questions stakeholders might ask and prepare additional details/data points to address them efficiently. When presenting problems, always propose possible solutions to demonstrate proactive understanding and facilitate decision-making. Use a clear summary or "TLDR" at the top of emails to highlight the main point or action required.

Project Closeout Report Purpose & Components

Good project hygiene: Compiling all links and documentation into one place for historical purposes. A **closeout report** compiles all project documentation, summaries, and links in one place, ensuring good project hygiene. It serves as a reference document for the organization, confirms the project is officially "done", reflects on team performance, confirms task completion, and serves as an invaluable reference for future projects and project managers.

Benefits of an effective closeout report:

- Helps ensure that everyone is satisfied with the work that was done.
- Finalizes the efforts of the team and lets people cleanly move on to new projects and tasks.
- Increases the impact of the team's work through strategic communication with other people across the organization who may not have been directly involved in the project.

A project closeout report includes:

1. **Project summary:** Detailed overview including core objectives, scope, and desired results.
2. **Methodology:** The approach or framework used to execute the project, such as Waterfall, Agile, or Lean.
3. **Performance baseline:** Comparing actual results to the originally planned schedule, cost, and scope, with detailed notes on any discrepancies.
4. **Outcomes:** Summarizes final deliverables, success metrics, feedback, and key accomplishments.
5. **Lessons learned:** Insights compiled from project retrospectives and reviews.
6. **Next steps:** Post-launch operations, ownership handoffs, or upcoming phases.
7. **Project documentation archive:** The compiled links and repository files for historical records.

The report helps finalize the project thoroughly and professionally, and it can also serve as an excellent portfolio piece for project managers to demonstrate growth and readiness to potential employers.

Purpose and Importance of Impact Reporting

Impact reports: Demonstrate the value added by a completed project, typically prepared for and presented directly to senior stakeholders, supporters, funders, or sponsors. Reporting your impact can help you to:

- Analyze results to adapt and improve services.
- Motivate staff and senior stakeholders through celebrating achievements.
- Build trust and credibility with supporters, sponsors, funders, and anyone benefiting from the project.
- Share lessons and insights with similar organizations.

Executive Summary Essentials & Drafting Process

Executive summary: A concise overview (ranging from a few sentences to a single comprehensive paragraph) highlighting the project's core purpose, vision, and final outcomes. It provides an overview of the main points of a larger report, written specifically to share with busy stakeholders who do not have time to review the entire document. It serves as a visual "highlight reel" of the project.

Your executive summary should aim to answer questions like: *How effectively was the project delivered? What did we learn from it?*

Process for Creating an Executive Summary:

1. Review foundational project initiation and planning documents like SMART goals, the business case, and the project charter to identify key accomplishments and outcomes.
2. Draft the detailed, full impact report first (including comprehensive results, what worked, and next steps).
3. Extract the most relevant, high-impact information from that report to write the summary.

Topics to include in the executive summary: Project vision, Key accomplishments (such as financial impact, milestones achieved, or major performance results), and Lessons learned for future improvements. *Example:* Highlighting a project that launched a money-saving app, detailing its vision, major financial accomplishments, and user feedback for future optimizations.

Reflecting on Accomplishments and Career Planning

- **Reflecting on Accomplishments and Lessons Learned:** Identify key accomplishments and challenges you overcame throughout the program, including mastering difficult concepts or balancing personal commitments. Reflect deeply on lessons learned, such as personal preferences for certain project management tasks or specific areas where you want to continuously improve. Celebrate your successes and strive to always keep growing and improving.
- **Planning Next Steps for Career Advancement:** Determine actionable next steps to advance your project management career, like networking, seeking more responsibilities, or applying for jobs. Organize these goals into a timeline, treating your personal career development as a project to manage. Use your personal closing report as a project artifact to showcase your growth and readiness to potential employers, celebrating your journey and skills gained. Take your personal closing report with you as a permanent project artifact to look back on your experience throughout this program.

Summary of the Capstone Core Competencies Covered

- **Project Initiation and Planning:** Created a project charter to define goals and deliverables. Conducted stakeholder analysis and planned tasks using team brainstorming and advanced time estimation techniques (including Effort vs. Duration, Three-Point Estimating, Beta/PERT distribution, and Confidence level ratings). Prepared for negotiations with stakeholders and learned how to influence using frameworks (Conger's steps) and temporary coalitions.
- **Project Execution and Quality Management:** Executed tasks while setting and measuring quality standards across standard categories using user surveys. Built comprehensive survey development processes (Evaluation questions, indicators, open/closed questions) and explored presentation reporting styles (Full reports, summary sheets, slides) driven by narrative storytelling. Facilitated productive, blameless team retrospectives to manage participation, separate accountability from blame, turn complaints into SMART action items, and navigate negativity.
- **Project Closing and Reporting:** Connected project problems to high-level goals using OKRs. Wrote problem summaries and best-practice escalation emails to stakeholders for quick decisions. Explored the role of problem-solving in program management and utilized generative AI for communication layouts. Completed a final closeout report ensuring good project hygiene, documented project impact reporting to build credibility, and mastered writing impactful executive summaries for senior stakeholders.

AI Application Scenarios Summary

- Stuck at the beginning of a project? You'll learn how to use AI tools to generate ideas and content.

- Planning an event? You'll use AI tools to help research, organize, and make more informed decisions.
- Drowning in a flooded inbox? You'll use AI tools to help speed up those daily work tasks like drafting email responses.